

Comptia A+ Assignment

Module -1, 2 [Hardware and its components]

GPUs and Cooling Systems

Basics of GPUs and cooling systems (air vs. liquid cooling). Identify a GPU and cooling system in a PC setup.

Task

- ❖ **1. Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.**
- A Graphics Processing Unit (GPU) is a hardware component responsible for processing graphics and displaying images, videos, animations, and games on a computer screen. GPUs are also used in tasks such as video editing, 3D rendering, artificial intelligence (AI), and machine learning.
- There are two main types of GPUs:
 - **Integrated GPU (iGPU):** Built into the processor (CPU). It shares system memory (RAM) and is suitable for everyday tasks such as web browsing, watching videos, and office work.

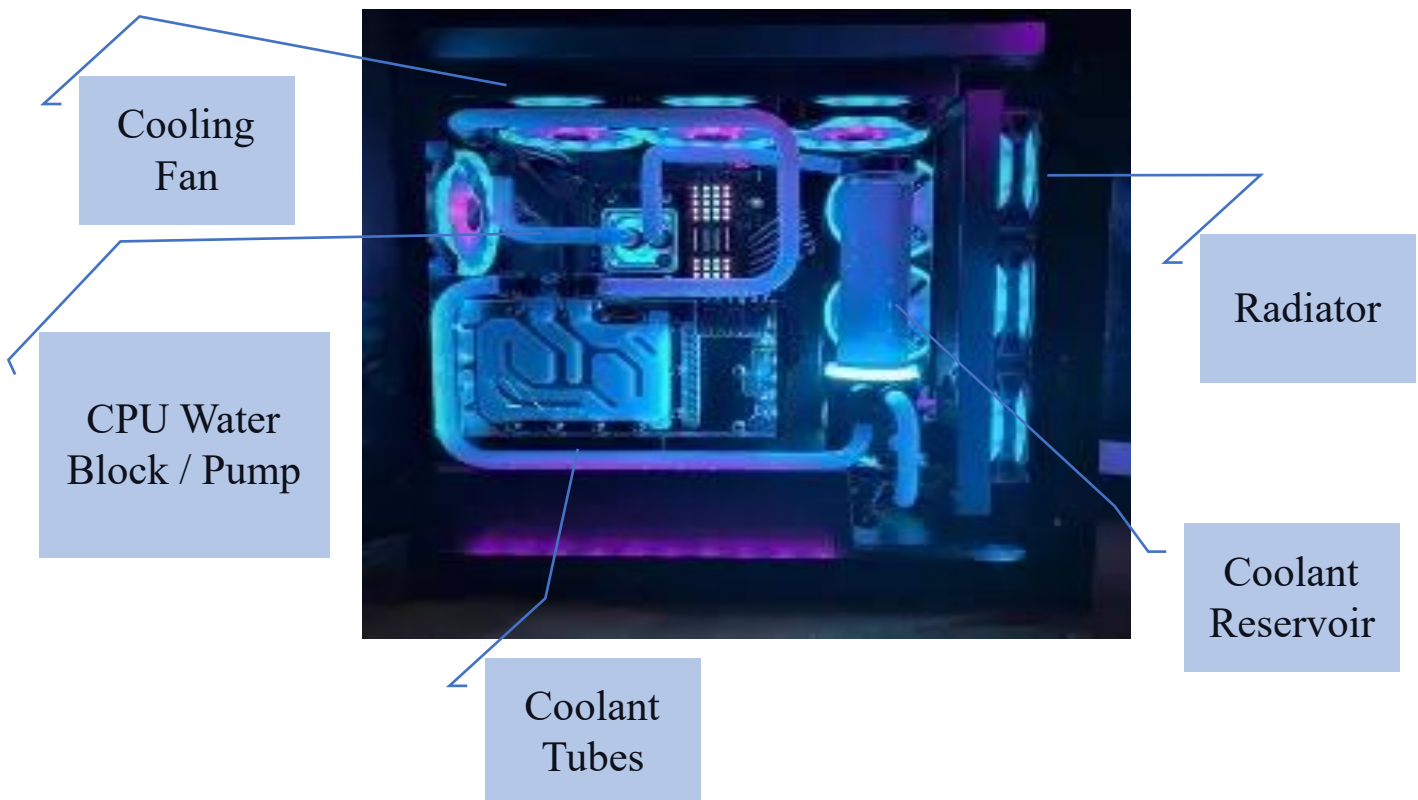
- **Dedicated GPU (dGPU):** A separate graphics card with its own memory (VRAM). It provides much higher performance and is mainly used for gaming, video editing, CAD, and professional graphics applications.

➤ Identified The GPU

- **Brand:** - Intel
- **Model Number:** - Intel UHD Graphics 620
- **GPU Type:** - Integrated GPU

- ❖ **2. Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function.**

➤ PC Liquid Cooling System



➤ Components and their function.

1. Cooling Fans

- The cooling fans blow air through the radiator.
- They help remove heat from the liquid coolant.

2. Radiator

- The radiator transfers heat from the coolant to the air.
- It keeps the liquid cool before it returns to the CPU.

3. CPU Water Block / Pump

- It is attached to the CPU.
- It absorbs heat from the processor and circulates the coolant.

4. Coolant Tubes

- These tubes carry the liquid coolant between the CPU, radiator, and reservoir.
- They allow the coolant to flow through the cooling system.

5. Coolant Reservoir

- It stores extra coolant.
- It helps maintain proper coolant flow and pressure.

❖ **3. Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.**

➤ Below given Comparison Between Air Cooling and Liquid Cooling Systems for GPUs.

Feature	Air Cooling	Liquid Cooling
Cost	Less expensive and budget-friendly.	More expensive due to additional components.
Efficiency	Provides good cooling for normal use and moderate gaming.	Offers better cooling performance, especially under heavy workloads and overclocking.
Maintenance	Requires very little maintenance; only occasional dust cleaning.	Requires more maintenance and regular inspection of the cooling system.

❖ **4. Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.**

- So, My Choice is Liquid Cooling.
- If I were building a gaming PC for streaming IPL matches and playing graphics-heavy games, I would choose liquid cooling for my GPU.
- Liquid cooling provides better heat dissipation than air cooling, which helps keep the GPU at a lower temperature during long gaming and streaming sessions.
- It also reduces the risk of overheating, improves system stability, and allows the GPU to maintain high performance under heavy workloads.
- Although liquid cooling is more expensive, its superior cooling efficiency and quieter operation make it the best choice for a high-performance gaming PC.